

CSCI-UA 9102 – Data Structures

Assignment 1

Dating App Profile System (20pts)

Summer 2026

Overview

In this assignment you will develop the core components of a simple dating application using Java. The purpose of this project is to practice:

- Object-Oriented Programming
- Classes and Objects
- ArrayLists
- Searching through collections
- Basic GUI programming using Java Swing

The application will allow users to create profiles, display profiles in a graphical user interface, and determine whether two users are compatible.

Learning Objectives

Upon successful completion of this assignment, students should be able to:

- Design a class with appropriate attributes and methods.
- Store objects in an `ArrayList`.
- Traverse collections using loops.
- Implement a simple matching algorithm.
- Create a graphical user interface using Java Swing.

Part 1: Create the Profile Class (5pts)

Create a class named `Profile`.

Required Instance Variables

Your class must contain at least the following attributes:

- Name
- Age
- Biography
- Collection of interests

You may add additional fields if desired.

Required Methods

Implement the following:

1. A constructor that initializes all instance variables.
2. Getter methods for all attributes.
3. A `toString()` method that returns a formatted description of the profile. This method should return an object of class `StringBuilder`. The `StringBuilder` class can be used to enclose a `String` using the method “`append`” (see documentation online). The string should look like the one given in Fig. 1.

Part 2: Display the Profile in a GUI (2pts)

Once your profile has been completed, you should be able to display it with the method `displayProfileGUI()`. The method implements a graphical user interface using Java Swing. Once the assignment will be completed, the GUI should display:

- A picture representing the user (in this case the representation is a drawing but you should feel free to modify)
- The description of the user.

Part 3: Matching Profiles (5pts)

Create a method `boolean isMatch(Profile other)` that should determine whether two profiles are compatible. You may define compatibility in any reasonable way. Examples include:

- Number of shared interests
- Similar age range
- Combination of multiple criteria

Or (preferred) any combination of those. Document your matching strategy in comments.

```

+-----+
| Picture |
+-----+
| Name: Alice |
| Age: 21 |
| |
| Bio: |
| Computer Science major |
| |
| Interests: |
| - Hiking |
| - Programming |
| - Travel |
+-----+

```

Figure 1: Example text that should be returned by the function `toString()`

Part 4: Create the DatingApp Driver Class (5pts)

Create a separate class named `DatingApp`. This class must contain the `main()` method.

Requirements

Within `main()`:

1. Create at least three profiles.
2. Store them in an `ArrayList`.
3. Print all profiles to the console.
4. Display at least one profile using the GUI.
5. Test the matching algorithm between profiles.

Additional features (3 Points)

Implement one of the following:

- Sort profiles by age.
- Search for profiles containing a specific interest.
- Display multiple profiles in separate windows.
- Add a “Like” system between users.

Submission Requirements

Submit a single file `DatingApp.java` on Brightspace. Ensure that:

- Your code compiles without errors.
- Your program runs successfully.
- The GUI displays correctly.
- Your code is properly commented.

Grading Rubric